Curated Results, Tables, and Figure Compendium

Companion to Manuscript v11 – DCT1-high parent-gene phosphosite enrichment in mouse spaceflight kidney

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Draft v11 results compendium, May 2026

This document collects the curated results, manuscript tables, figure gallery, and artifact index supporting Manuscript v11. It is the technical supplement for the streamlined manuscript: cross-cohort RNA recurrence, matched OSD-462 RNA/protein/phosphoprotein mismatch, DCT1-high parent-gene phosphosite enrichment, parent-protein and compartment-score sensitivity models, exploratory path analysis, external spatial-reference context, and the perturbation-triangulation pass (low-K DCT reference, parent-protein-normalized phosphosite effects, IRI DCT-transport spatial context, dDAVP/mpkDCT, KLHL3/CUL3 turnover). The main manuscript carries the story; this compendium carries the extra tables, secondary analyses, and run artifacts.

Contents

S1 Run provenance and executable entrypoint

Table S1: **V11 run provenance.**

Run root	`data/results/run_20260526_v11_dct1_phospho_mediation/`
Plan	`docs/v11_execution_research_plan.md`
Execution report	`docs/v11_execution_results.md`
Plan checksum	`d8b7465e09257dd4444fc2abb969a02066f2d360e4fbc04056791a60c9755e3f`
Pipeline entrypoint	`python -m src.run_all_phases --v11-only --run-id <run_id>`
Canonical v11 modules	`src/v11/core_analysis.py`; `src/v11/h2_composition_aware_phospho.py`; `src/v11/spatial_reference_projection.py`; `src/v11/publication_figures.py`; `src/v11/build_gse228367_dct_prior.R`

Table S2: **External datasets added or reused for v11.**

Layer	Dataset	V11 role
Spaceflight RNA recurrence	RRRM-2 / OSD-771, OSD-513, OSD-462	Defines and tests the recurrent matrix/endothelial-high and DCT-low RNA state.
Spaceflight multi-omics	OSD-462 / RR-10	Whole-kidney matched RNA, protein, and phosphoprotein anchor.
DCT subtype prior	GSE228367	Native kidney DCT1/DCT2 snRNA-seq reference used to score parent-gene DCT1 enrichment.
DCT-lineage phosphoproteome	PXD001729	mpkDCT dDAVP phosphoproteomic reference; used for DCT-lineage plausibility and negative anti-alignment boundary.
Spatial reference	GSE269622	Whole-transcriptome Visium IRI timecourse used for pathway-vector projection.
Spatial annotation context	GSE269719	Targeted 299-gene Xenium object used for cell-type and cellular-neighborhood inventory.

S2 Executive verdict and claim labels

Table S3: **Main result verdicts.**

Hypothesis	Question	Verdict
RNA recurrence	Does a recurrent matrix/endothelial-high and DCT-low RNA remodeling state recur across cohorts and in OSD-462 RNA?	Supported; remains the backbone.
DCT1-high enrichment	Are OSD-462 flight-suppressed phosphosites enriched among DCT1-high parent genes?	Supported as top-decile subset enrichment.
Sensitivity models	Does the enrichment remain after parent-protein and compartment-score adjustment?	Top-decile subset survives; continuous gradient does not.
Exploratory path analysis	Are remodeling scores statistically related to NCC/SPAK phospho suppression?	Endothelial and matrix/endothelial paths are directionally consistent; temporal order remains experimental.
Spatial supplement	Which injury/repair spatial niches resemble the bulk spaceflight RNA vector?	Contextualized to repair-stage, DCT-adjacent/injured-tubule and fibro-inflammatory/endothelial niches.
Perturbation triangulation	Do public perturbation references (low-K DCT, parent-protein normalization, IRI DCT-transport, dDAVP, KLHL3/CUL3) identify the upstream WNK-suppressing mechanism?	No single upstream signal identified; parent-protein-normalized DCT1 enrichment is a robustness upgrade; IRI Visium DCT-adjacent transport drop is a testable spatial prediction.

Table S4: **Preferred and avoided manuscript language.**

Topic	Preferred wording	Avoid
Overall frame	DCT1-high parent-gene phosphosite enrichment within recurrent kidney remodeling.	DCT1-anchored mechanism without qualification.
DCT1 enrichment	DCT1-high top-decile enrichment of flight-suppressed whole-kidney phosphosites.	DCT1-specific phosphoproteomics.
Composition models	Conservative sensitivity adjustment for parent protein abundance and estimated bulk compartments.	Fully deconfounded or deconvolved cell-of-origin analysis.
Path analysis	Consistent with a negative indirect path through endothelial or matrix/endothelial remodeling.	Endothelial remodeling causes NCC dephosphorylation.
Spatial	External IRI spatial reference contextualization.	Spatial validation of the spaceflight lesion.

S3 RNA recurrence and RNA-protein mismatch

Table S5: **Core RNA recurrence and multi-omic quantities from the v11 run.**

Component	Value	Interpretation
OSD-462 RNA recurrence	0.869	OSD-462 recurs the RRRM-2 matrix-high / DCT-low direction.
OSD-462 protein abundance	null	RNA remodeling does not propagate cleanly as protein abundance.
NCC regulatory phospho score	-0.832	NCC regulatory phosphosites are suppressed while total NCC protein is flat.
Targeted KSEA SPAK/OSR1	-6.31	Inferred kinase-output down; three quantified curated substrates.
Targeted KSEA WNK	-4.12	Inferred WNK-output down; three quantified curated substrates, interpreted as pathway coherence.
Endothelial score vs NCC phospho	-0.762	Strongest compartment anti-correlation with NCC regulatory phosphorylation.

Interpretation: recurrent RNA remodeling is strong, targeted protein-abundance concordance is null, and the transporter lesion resolves at the phospho/activity layer.

S4 GSE228367 DCT subtype prior

Table S6: **GSE228367 DCT reference inventory.**

DCT1 nuclei	18,881
DCT2 nuclei	6,274
Normal-potassium replicates per subtype	3
Strict FDR DCT1 core genes	2
Strict FDR DCT2 core genes	0

Table S7: **Selected marker sanity checks in the DCT1/DCT2 reference.**

Gene	DCT1 mean	DCT2 mean	Read
Slc12a3	5.14	4.40	present in both, higher in DCT1
Pvalb	0.279	0.024	DCT1-high
Trpm6	2.40	1.90	DCT1-high
Klhl3	2.84	2.65	mild DCT1-high
Trpv5	0.006	0.711	DCT2-high
Calb1	1.01	2.97	DCT2-high
Stk39	1.98	1.54	DCT1-leaning
Wnk4	1.72	1.40	DCT1-leaning

Interpretation: the strict FDR marker sets are too sparse for binary enrichment. V11 therefore uses a continuous DCT1 enrichment score and DCT1-high percentile bins.

S5 DCT1-high parent-gene phosphosite enrichment

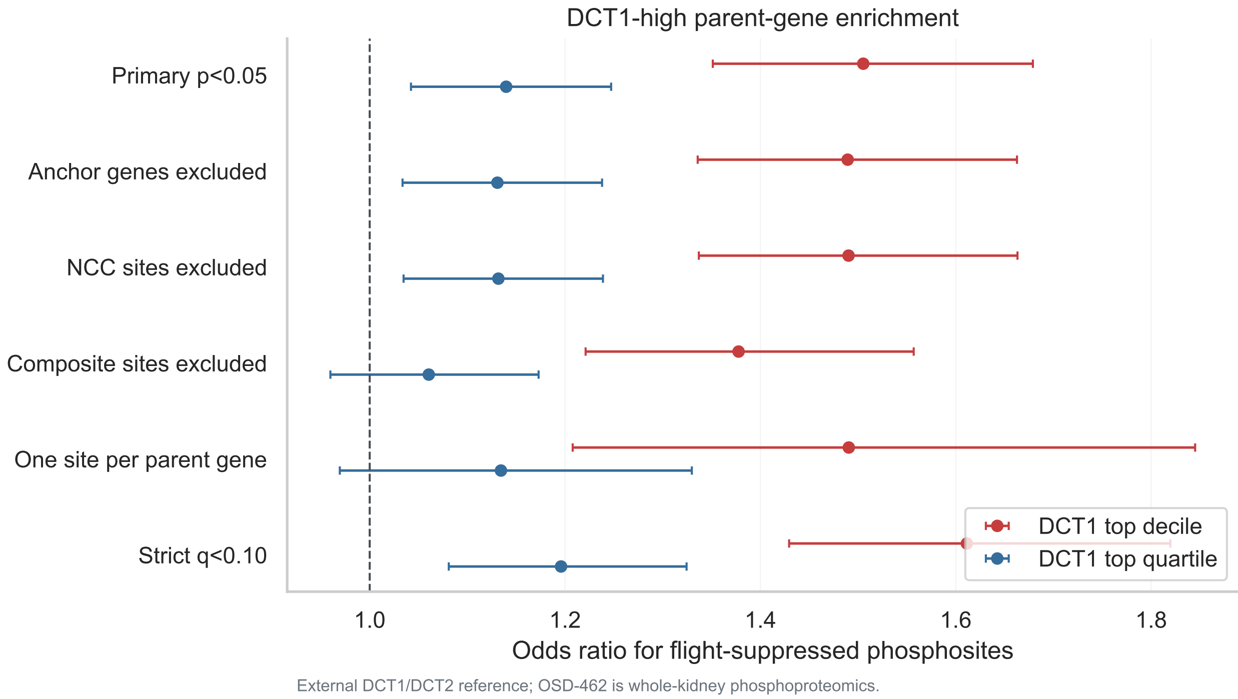

Figure S1: **DCT1-high parent-gene enrichment.** Top-decile and top-quartile DCT1 reference bins are tested across primary and sensitivity definitions. The strong and stable signal is the DCT1 top decile.

Table S8: **Primary DCT1-high enrichment tests.**

Test	Suppressed set	Result	q	Read
Matched-null mean DCT1 score	p¡0.05 suppressed sites	–	0.063	weak continuous support at FDR 0.10
Fisher DCT1 top quartile	p¡0.05 suppressed sites	OR 1.14	0.0062	supported but weaker than top decile
Fisher DCT1 top decile	p¡0.05 suppressed sites	OR 1.51	9.96×10^{-12}	supported at row level
DCT1 top decile, anchor genes excluded	p¡0.05 suppressed sites	OR 1.49	2.77×10^{-11}	survives anchor exclusion
DCT1 top decile, NCC sites excluded	p¡0.05 suppressed sites	OR 1.49	2.77×10^{-11}	survives NCC-site exclusion
Composite/multi-position sites excluded	p¡0.05 suppressed sites	OR 1.38	1.86×10^{-6}	removes composite rows; not a parent-gene dependence fix
One site per parent gene	p¡0.05 suppressed sites	OR 1.49	6.40×10^{-4}	survives true parent-gene representative-site sensitivity
Parent-gene Fisher	parent has at least one p¡0.05 suppressed site	OR 1.52	1.52×10^{-4}	parent-gene-level confirmation
Parent-gene logistic	parent has at least one p¡0.05 suppressed site	OR 1.14	0.453	attenuated after site-count, peptide-count, and abundance covariates
Site-count-stratified permutation	row-level p¡0.05 suppressed sites	OR 1.51	$p_{\mathrm{emp}} = 0.0005$	DCT1 labels shuffled within parent-gene site-count bins

Interpretation: suppressed whole-kidney OSD-462 phosphosites are enriched among parent genes that are DCT1-high in the external DCT reference. Direct cell-of-origin assignment would require DCT-enriched or spatially resolved spaceflight phosphoproteomics.

S6 Parent-protein and composition-score sensitivity

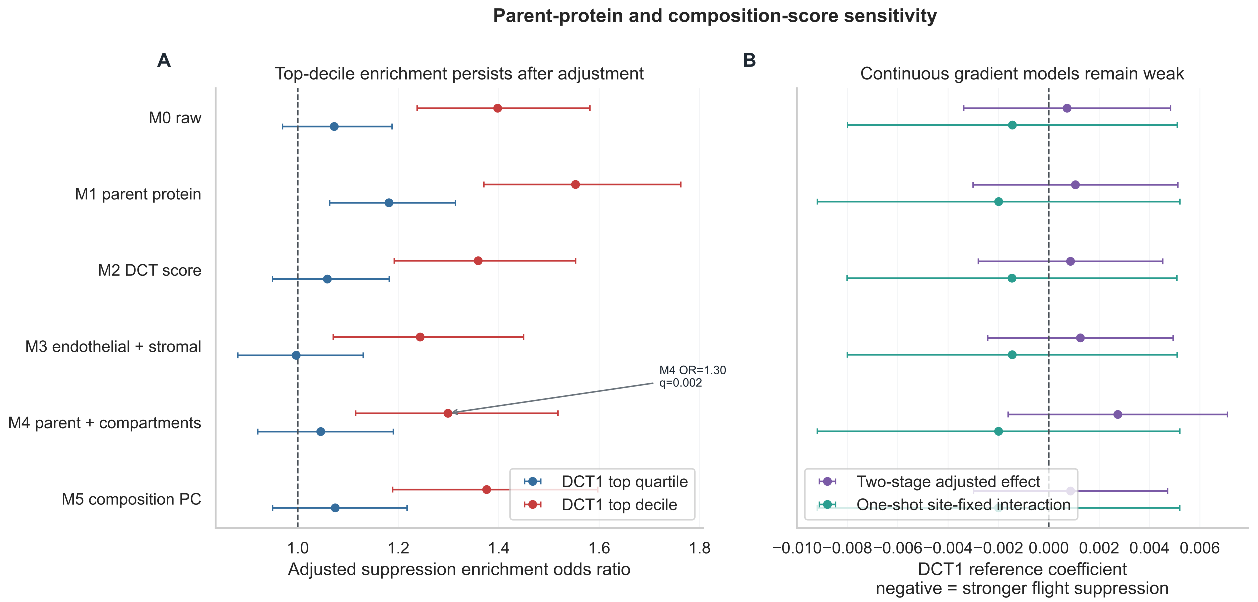

Figure S2: **Parent-protein and composition-score sensitivity.** Panel A shows top-decile and top-quartile enrichment across the adjustment ladder. Panel B shows that continuous DCT1-reference and flight-by-DCT1 interaction models remain weak.

Table S9: **DCT1 top-decile adjusted-suppression enrichment across the composition-aware model ladder.**

Model	Adjustment	OR	q	Read
M0	Raw flight effect	1.40	6.35e-7	raw DCT1-high subset enrichment
M1	Parent protein abundance	1.55	2.14e-10	not explained by parent protein abundance
M2	DCT score	1.36	2.17e-5	survives estimated DCT abundance adjustment
M3	Endothelial + stromal scores	1.24	0.00545	attenuated, still positive
M4	Parent protein + DCT + endothelial + stromal	1.30	0.00158	full model remains positive
M5	Parent protein + composition PC1	1.38	6.53e-5	PC-based composition adjustment passes

Table S10: **Continuous DCT1-reference models do not pass.**

Test	Key term	Coefficient	q	Read
Two-stage M0 DCT1-only	DCT1 prior	-0.00110	0.877	weak negative, not significant
Two-stage M4 full	DCT1 prior	+0.00274	0.877	no continuous suppression gradient
One-shot LM0 site-fixed	flight x DCT1 prior	-0.00145	0.665	direction negative, weak
One-shot LM4 full site-fixed	flight x DCT1 prior	-0.00200	0.665	direction negative, weak

Interpretation: the robust result is top-decile subset enrichment after parent-protein and compartment-score adjustment, not a broad continuous DCT1-gradient effect.

S7 PXD001729 and KLHL3/CUL3 checks

Table S11: **PXD001729 dDAVP anti-alignment check.**

Comparison	Shared sites	Shared genes	Cosine	Read
All shared single sites	60	49	-0.026	essentially null / unstable; 95% bootstrap CI -0.310 to 0.287
Transport target shared sites	0	0	NA	not testable

Table S12: **KLHL3/CUL3 and WNK/SPAK/NCC coverage.**

Gene	OSD-462 phosphosites	PXD phosphosites	Protein effect	Read
Klhl3	0	0	+0.013	no phosphosite evidence
Cul3	1	0	-0.001	no useful turnover signal
Wnk1	12	12	-0.024	WNK1 phospho observed, protein mostly flat
Wnk4	8	1	+0.087	WNK4 phospho observed, protein not depleted
Stk39/SPAK	5	3	+0.081	phospho suppression despite protein not depleted
Slc12a3/NCC	16	0	+0.089	phospho suppression with protein flat/up

Interpretation: PXD001729 supports DCT-lineage phosphoproteome plausibility but not targeted vasopressin anti-alignment. KLHL3/CUL3 turnover remains a candidate future mechanism because public data lack ubiquitinomics and KLHL3 phosphosite coverage.

S8 Exploratory path analysis

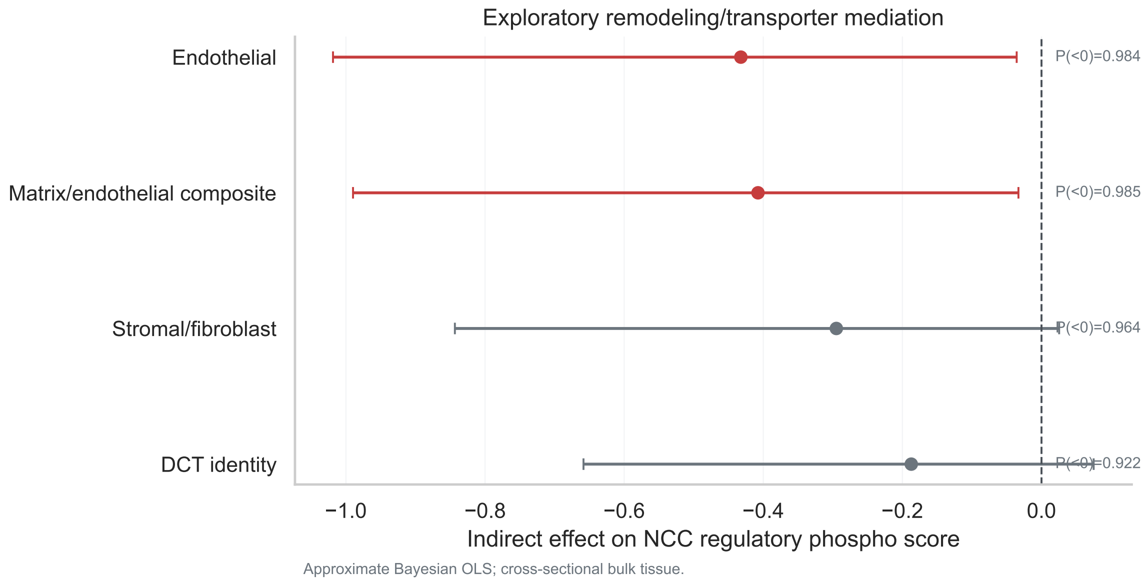

Figure S3: **Exploratory remodeling/transporter path model.** Endothelial and matrix/endothelial composite indirect paths are negative with intervals excluding zero. Cross-sectional bulk data prevent causal interpretation.

Table S13: **Indirect path summaries from approximate Bayesian OLS path models.**

Mediator	Indirect median	95% low	95% high	Read
Endothelial	-0.432	-1.019	-0.036	consistent with negative indirect path
Stromal/fibroblast	-0.295	-0.843	0.025	suggestive, interval crosses zero
DCT identity	-0.187	-0.658	0.075	composition-linked, interval crosses zero
Matrix/endothelial composite	-0.408	-0.990	-0.033	consistent with negative indirect path

Table S14: **Approximate future-n planning from observed effect scale.**

Mediator	Approximate n for 80% interval-exclusion power
Endothelial	40 animals
Matrix/endothelial composite	40 animals
Stromal/fibroblast	60 animals
DCT identity	80 animals

Interpretation: the data are consistent with a remodeling-linked path or shared remodeling, not causal mediation. The direct flight effect remains negative after intermediate-score adjustment.

S9 External spatial reference contextualization

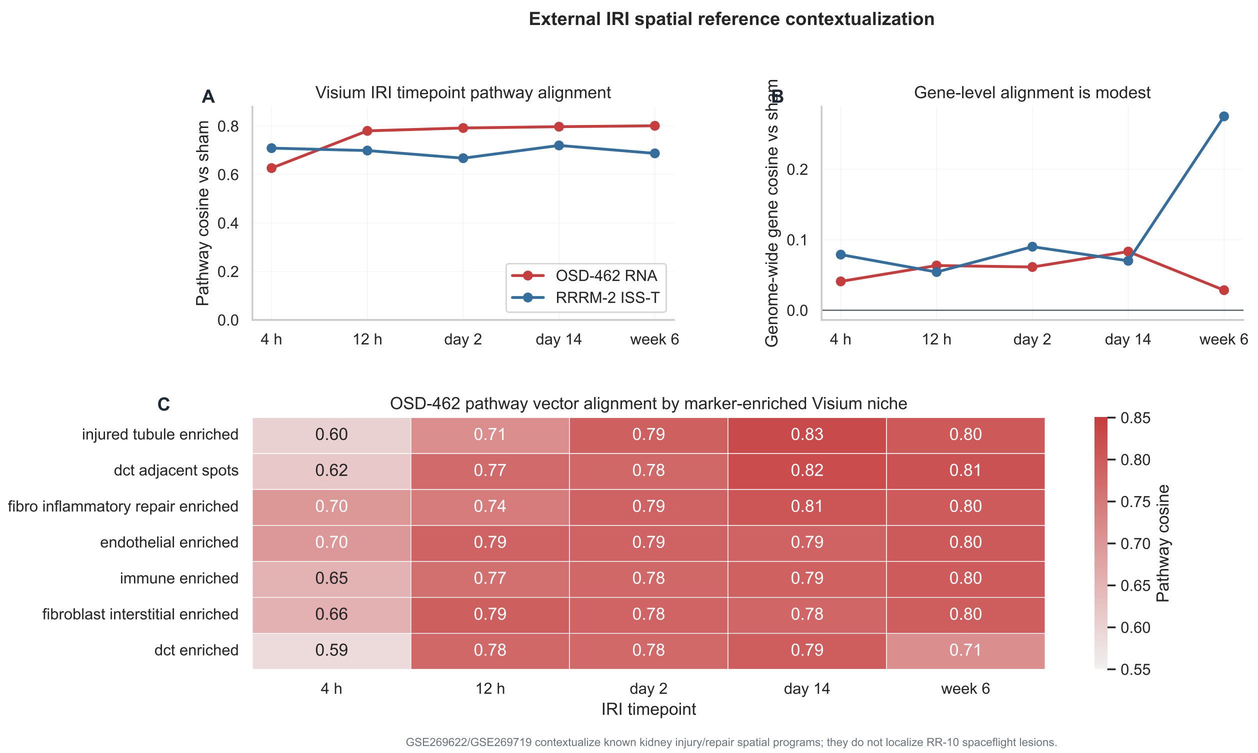

Figure S4: **External IRI spatial reference projection.** The OSD-462 RNA vector aligns strongly with Visium IRI repair-stage pathway programs. Spatial-niche matches are strongest in injured tubule, DCT-adjacent, fibro-inflammatory, and endothelial contexts.

Table S15: **Visium IRI timepoint cosines vs sham.**

IRI timepoint	OSD-462 pathway	RRRM-2 pathway	OSD-462 gene	RRRM-2 gene
4 h	0.627	0.709	0.041	0.079
12 h	0.780	0.699	0.063	0.054
day 2	0.792	0.667	0.061	0.090
day 14	0.797	0.720	0.083	0.070
week 6	0.801	0.687	0.028	0.275

Table S16: **Top OSD-462 Visium niche pathway matches.**

Timepoint	Niche	Cosine
day 14	injured tubule-enriched spots	0.827
day 14	DCT-adjacent spots	0.825
week 6	DCT-adjacent spots	0.814
day 14	fibro-inflammatory repair-enriched spots	0.807
week 6	injured tubule-enriched spots	0.802
week 6	endothelial-enriched spots	0.802

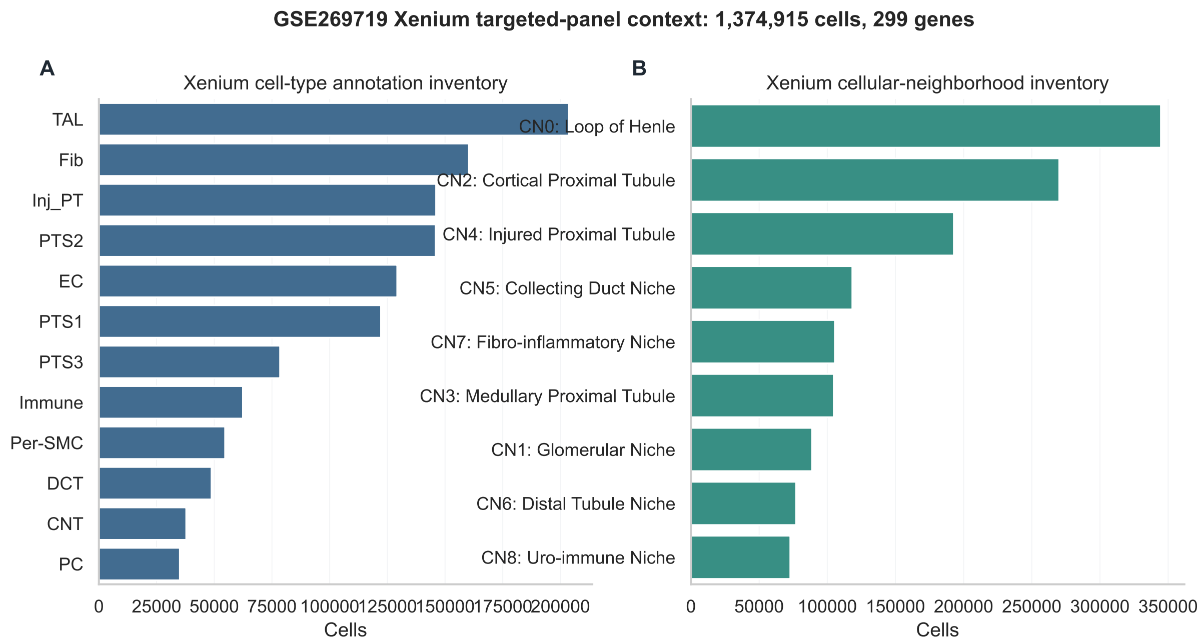

Figure S5: **GSE269719 Xenium annotation inventory.** The Xenium object provides targeted-panel annotation and cellular-neighborhood context, not whole-transcriptome pathway projection.

Table S17: **Selected Xenium annotation counts.**

Annotation	Count	Note
Total cells	1,374,915	processed AnnData object loaded in backed mode
Genes	299	targeted panel, not genome-wide
DCT cells	48,774	`celltype\_plot` annotation
Endothelial cells	129,261	`celltype\_plot` annotation
Fibroblasts	160,328	`celltype\_plot` annotation
Injured proximal tubule cells	146,097	`celltype\_plot` annotation
Distal tubule niche cells	77,103	`CN` annotation
Fibro-inflammatory niche cells	105,536	`CN` annotation

Interpretation: spatial evidence is a contextual supplement. It supports the wording "spatially contextualized," not "spatially validated."

S10 Perturbation triangulation

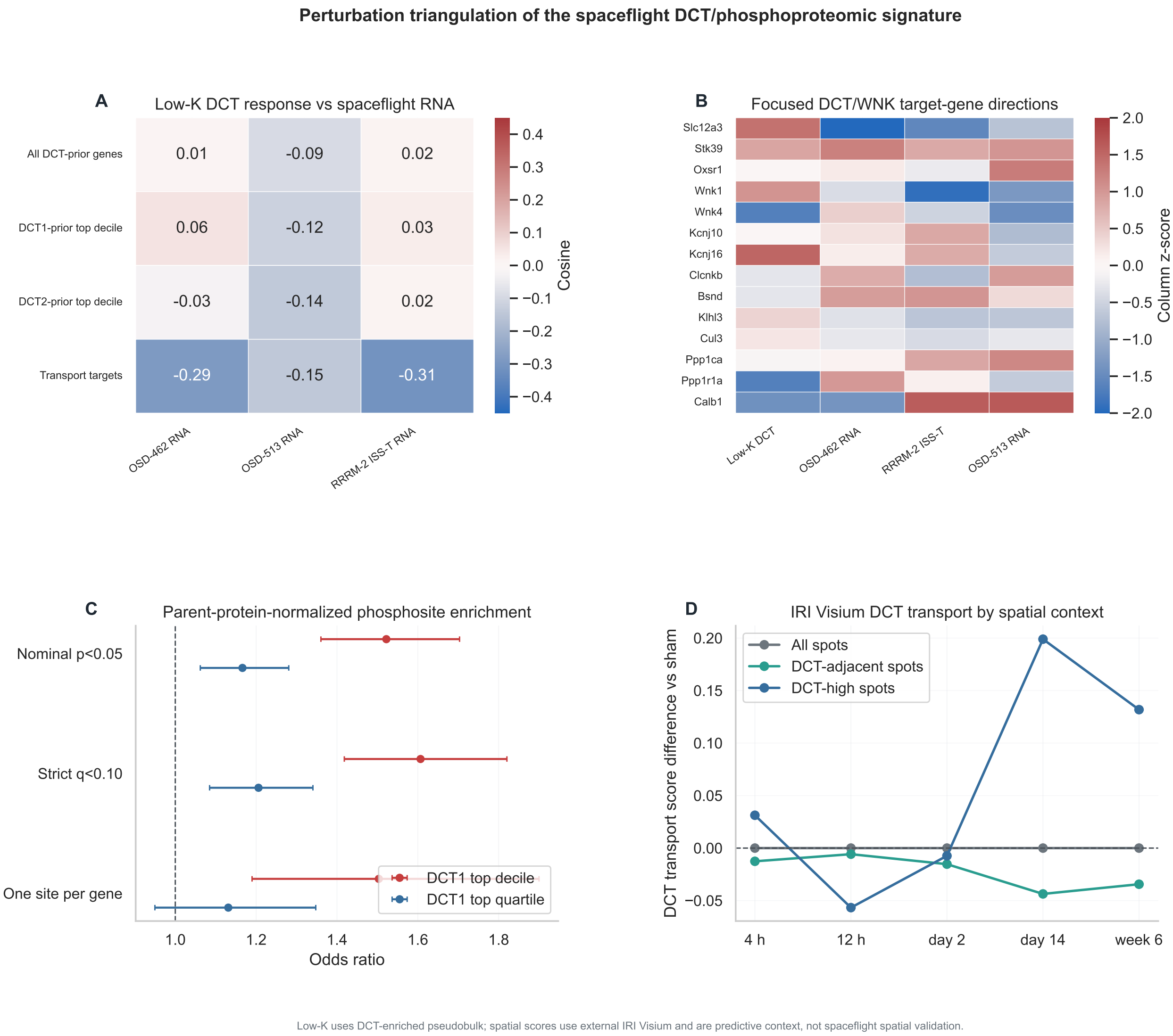

Figure S6: **Perturbation triangulation.** (A) Low-K DCT response cosines with OSD-462, OSD-513, and RRRM-2 ISS-T RNA by gene-prior class. (B) Focused DCT/WNK transport-target gene direction heatmap. (C) Parent-protein-normalized DCT1 enrichment ORs. (D) IRI Visium DCT-transport score by timepoint and spatial context.

Table S18: **Perturbation-triangulation branch verdicts.**

Mechanism branch	Reference and analysis	Headline result	Paper role
Potassium/chloride-WNK	GSE228367 low-K DCT-enriched pseudobulk anti-alignment	Directional anti-alignment in 14-gene transport-target subset (cosine −0.286 OSD-462, −0.311 RRRM-2 ISS-T, −0.146 OSD-513), wide bootstrap CIs, did not meet promotion rule.	Mechanism triage, not primary upstream mechanism.
Parent-protein normalization	OSD-462 parent-protein-normalized phosphosite re-test	DCT1 top-decile OR 1.52 (CI 1.36–1.70, q=3.73×10^{-12}); survives anchor, NCC, one-site-per-parent-gene, and strict-q sensitivities.	Robustness upgrade to the DCT1-high parent-gene result; not direct biochemical occupancy.
Injury/repair spatial context	GSE269622 IRI Visium DCT-transport score by spatial niche	DCT-adjacent spots drop at day 14 (−0.044, $p = 0.005$) and week 6 (−0.034, $p = 0.018$); DCT-marker-high spots rise at the same timepoints.	Spatial prediction/context for future spaceflight kidney sections.
Vasopressin/cAMP	PXD001729 mpkDCT dDAVP	60 shared single sites, 0 shared transport-target sites; key DCT/WNK targets do not map.	Secondary negative; coverage-bounded.
KLHL3/CUL3 turnover	OSD-462 KLHL3/CUL3/WNK coverage	0 *Klhl3* phosphosites, 1 *Cul3* site, flat WNK/SPAK/NCC protein abundance; no public kidney ubiquitinomics.	Future-data rationale.

S10.1 Branch 1: GSE228367 low-K DCT anti-alignment

Table S19: **Low-K DCT response vs spaceflight RNA, 14-gene transport-target subset.**

Spaceflight vector	n genes	Cosine	95% CI	Spearman ρ	p	q
OSD-462 RNA	14	−0.286	−0.752 to 0.458	−0.420	0.135	0.218
RRRM-2 ISS-T RNA	14	−0.311	−0.780 to 0.325	−0.341	0.233	0.306
OSD-513 RNA stat	14	−0.146	−0.768 to 0.506	−0.169	0.563	0.695

Interpretation: the 14-gene transport-target subset moves directionally opposite the canonical low-K DCT activation direction in all three spaceflight cohorts (10 of 14 focused DCT/WNK genes flip in OSD-462, including *Slc12a3*, *Wnk1*, *Klhl3*, *Cul3*), but none of the comparisons clears the pre-specified primary threshold (cosine < -0.3 with CI excluding zero). Promotion rule not met; treated as mechanism triage.

S10.2 Branch 2: parent-protein-normalized DCT1 enrichment

Table S20: **Parent-protein-normalized phosphosite enrichment among DCT1-high parent genes.** Parent-normalized effect = phosphosite flight effect minus parent-protein flight effect. This is a robustness summary, not direct biochemical occupancy.

Analysis	DCT1 subset	OR	95% CI	q	Read
parent-normalized $p < 0.05$	top decile	1.52	1.36–1.70	3.73×10^{-12}	robustness upgrade holds
parent-normalized $p < 0.05$	top quartile	1.17	1.06–1.28	0.00113	supported
parent-normalized strict $q < 0.10$	top decile	1.61	1.42–1.82	3.73×10^{-12}	stricter set, OR rises
parent-normalized strict $q < 0.10$	top quartile	1.21	1.08–1.34	6.83×10^{-4}	supported
exclude anchor genes	top decile	1.51	1.34–1.69	9.75×10^{-12}	survives anchor exclusion
exclude NCC sites	top decile	1.51	1.35–1.69	9.75×10^{-12}	survives NCC exclusion
one site per gene	top decile	1.50	1.19–1.90	8.42×10^{-4}	survives parent-gene representative-site sensitivity

Table S21: **Selected target-site parent-protein-normalized examples.**

Gene	Site	Raw phospho effect	Parent protein effect	Parent-normalized effect
Slc12a3/NCC	p53	−0.851	+0.089	−0.940
Slc12a3/NCC	p65	−0.790	+0.089	−0.879
Slc12a3/NCC	p68	−0.794	+0.089	−0.883
Slc12a3/NCC	p89	−0.930	+0.089	−1.019
Stk39/SPAK	p366	−0.520	+0.081	−0.601
Stk39/SPAK	p383	−0.793	+0.081	−0.874

Interpretation: per-site parent-protein normalization sharpens the suppressed-site effect because parent NCC and SPAK protein effects are slightly positive. The DCT1-high enrichment is not explained by parent protein abundance in this sensitivity; the calculation remains a whole-kidney parent-normalized effect rather than direct phosphosite occupancy or biochemical stoichiometry.

S10.3 Branch 3: IRI Visium DCT-transport spatial context

Table S22: **Visium DCT-transport score versus sham, by spatial group and timepoint.** Spot-level descriptive two-sample t; no animal-level replication.

Spatial group	Timepoint	Mean diff vs sham	t	p	n spots vs sham
DCT-adjacent spots	day 14	−0.044	−2.81	0.0050	932 vs 1,134
DCT-adjacent spots	week 6	−0.034	−2.36	0.0182	1,574 vs 1,134
DCT-marker-high spots (top quartile)	day 14	+0.199	+8.02	2.25×10^{-15}	693 vs 715
DCT-marker-high spots (top quartile)	week 6	+0.132	+5.98	2.70×10^{-9}	979 vs 715

Interpretation: DCT-adjacent neighborhoods carry a DCT-transport drop in late IRI repair while DCT-marker-high spots retain or even increase the DCT-transport program. This is the spatial prediction that a future spaceflight kidney spatial experiment would test directly.

S10.4 Branch 4: PXD001729 dDAVP/mpkDCT

Table S23: **PXD001729 mpkDCT dDAVP reference vs OSD-462.**

Quantity	Value	Note
Shared single phosphosites	60	cross-platform/cross-species mapping
Shared changed transport-target sites	0	*Slc12a3*, *Stk39*, *Oxsr1*, *Wnk1*, *Wnk4*, *Klhl3*, *Cul3* not shared
All-shared-site cosine	−0.026	95% bootstrap CI −0.310 to 0.287

Interpretation: the cultured DCT-lineage dDAVP reference is coverage-bounded for the OSD-462 transport-target lesion and cannot test a vasopressin/cAMP anti-alignment hypothesis. Used as plausibility/coverage boundary only.

S10.5 Branch 5: KLHL3/CUL3/WNK turnover

Table S24: **KLHL3/CUL3/WNK coverage in OSD-462 and PXD001729.**

Gene	OSD-462 phosphosites	PXD phosphosites	OSD-462 protein effect	Read
Klhl3	0	0	+0.013	no phosphosite evidence
Cul3	1	0	−0.001	no turnover signal
Wnk1	12	12	−0.024	phospho observed, protein flat
Wnk4	8	1	+0.087	phospho observed, protein not depleted
Stk39/SPAK	5	3	+0.081	phospho suppressed despite flat protein
Slc12a3/NCC	16	0	+0.089	phospho suppressed with protein flat/up

Interpretation: public data cannot distinguish KLHL3/CUL3-mediated WNK turnover from ionic/osmotic suppression. The next experiment requires DCT-enriched or targeted ubiquitinomics and direct WNK protein turnover assays.

S11 Complete artifact index

Table S25: **V11 generated result artifacts indexed by source path.**

Category	Source artifact(s)
Run report	`docs/v11_execution_research_plan.md`; `docs/v11_execution_results.md`
V11 source modules	`src/v11/build_gse228367_dct_prior.R`; `src/v11/core_analysis.py`; `src/v11/h2_composition_aware_phospho.py`; `src/v11/spatial_reference_projection.py`; `src/v11/publication_figures.py`; `src/v11/perturbation_gse228367_lowk.py`; `src/v11/h2_occupancy_normalized_phospho.py`; `src/v11/spatial_dct_transport_check.py`; `src/v11/perturbation_triage_summary.py`
Runner integration	`src/run_all_phases.py` phase 11; `--v11`; `--v11-only`; `--v11-site-scope`; `--v11-skip-spatial`; `--v11-skip-visium`; `--v11-skip-xenium`; `--v11-skip-figures`
Baseline lock	`data/results/run_20260526_v11_dct1_phospho_mediation/baseline/v11_baseline_lock_summary.tsv`; `baseline/v11_baseline_input_manifest.json`
DCT reference prior	`dct_prior/gse228367_gene_stats_overall.tsv`; `dct_prior/gse228367_gene_stats_by_rep.tsv`; `dct_prior/gse228367_dct1_vs_dct2_de.tsv`; `dct_prior/gse228367_dct_prior_summary.json`

Category	Source artifact(s)
External QC	external_qc/gse228367_object_inventory.tsv; external_qc/gse228367_marker_qc.tsv; external_qc/pxd001729_table_inventory.tsv; external_qc/pxd001729_phosphosite_effects.tsv
DCT1 enrichment	h2_enrichment/h2_dct1_sensitivity_summary.tsv; h2_enrichment/h2_dct1_parent_gene_level_summary.tsv; h2_enrichment/h2_dct1_site_count_stratified_permutation.tsv; h2_enrichment/h2_dct1_enrichment_verdict.json
PXD001729 checks	h2_pxd/h2_pxd001729_ddavp_antialignment_summary.tsv; h2_pxd/h2_pxd001729_shared_sites.tsv; h2_pxd/h2_pxd001729_verdict.json
KLHL3/CUL3 checks	h2_klhl3/h2_klhl3_cul3_effects.tsv; h2_klhl3/h2_klhl3_cul3_verdict.json
Composition-aware models	h2_composition_adjusted/h2_composition_adjusted_suppression_enrichment_single.tsv; h2_composition_adjusted/h2_composition_effect_level_dct1_ladder_single.tsv; h2_composition_adjusted/h2_composition_site_fixed_interaction_ladder_single.tsv; h2_composition_adjusted/h2_composition_aware_verdict_single.json; h2_composition_adjusted/h2_composition_manifest_single.json
Exploratory path analysis	h3_mediation/h3_mediation_model_summary.tsv; h3_mediation/h3_mediation_posterior_draws.tsv.gz; h3_mediation/h3_mediation_power_simulation.tsv; h3_mediation/h3_mediation_verdict.json
Spatial reference	spatial_reference/spatial_reference_projection_verdict.json; spatial_reference/visium_timepoint_gene_cosines.tsv; spatial_reference/visium_timepoint_pathway_cosines.tsv; spatial_reference/visium_niche_cosines.tsv; spatial_reference/visium_spot_niche_counts.tsv; spatial_reference/xenium_annotation_inventory.tsv; spatial_reference/visium_dct_transport_by_niche.tsv; spatial_reference/visium_dct_transport_timepoint_summary.tsv; spatial_reference/visium_dct_transport_spot_scores.tsv.gz; spatial_reference/visium_dct_transport_verdict.json
Perturbation triangulation	perturbation/lowk_dct_pseudobulk_effects.tsv; perturbation/lowk_dct_alignment_summary.tsv; perturbation/lowk_target_gene_table.tsv; perturbation/lowk_dct1_dct2_specificity.tsv; perturbation/lowk_alignment_verdict.json; perturbation/perturbation_triage_summary.tsv; perturbation/perturbation_triage_verdict.json
Parent-protein-normalized re-test	h2_occupancy/h2_occupancy_site_effects.tsv; h2_occupancy/h2_occupancy_dct1_enrichment.tsv; h2_occupancy/h2_occupancy_target_sites.tsv; h2_occupancy/h2_occupancy_verdict.json
V11 figures	figures/v11/v11_dct1_parent_gene_enrichment.pdf; figures/v11/v11_parent_protein_composition_sensitivity.pdf; figures/v11/v11_exploratory_mediation_forest.pdf; figures/v11/v11_spatial_reference_projection.pdf; figures/v11/v11_xenium_annotation_inventory.pdf; figures/v11/v11_perturbation_triangulation.pdf
Downloaded spatial data	data/external/spatial_reference/GSE269622_Visium/*.tar.gz; data/external/spatial_reference/GSE269719_Xenium/Xenium.h5ad

S12 Verification notes

- The v11 figure module generated six PNG/PDF figure sets under `figures/v11/` (DCT1 parent-gene enrichment, parent-protein composition sensitivity, exploratory mediation forest, spatial-reference projection, Xenium annotation inventory, perturbation triangulation).
- The direct v11 figure generator and integrated `src.visualization.publication_figures` path both detect and regenerate v11 figures.
- Python syntax checks passed for the v11 core, composition-aware, spatial-reference, figure, perturbation low-K, parent-protein-normalized phosphosite, spatial DCT-transport, and triage modules, plus the runner and publication-figure integration.
- The runner dry-run passed for `python -m src.run_all_phases --v11-only --dry-run --run-id run\_20260526\_v11\_dryrun --v11-skip-spatial` and for the perturbation-triangulation modules.
- The GSE269719 Xenium MD5 matched the figshare record: `e6f5728ab7e47b499149f63b54e892c7`.
- The spatial scripts completed after loading six GSE269622 Visium archives and the processed GSE269719 Xenium AnnData object; the DCT-transport spatial check used the same Visium archives.
- The perturbation low-K analysis loaded six GSE228367 filtered 10x matrices (NK1–3 and KD1–3) and emitted alignment and target-gene tables under `perturbation/`.

Curated results compendium for Manuscript v11. This document should travel with `manuscript_ v11. tex` and `manuscript_ v11. pdf`. The manuscript result is a DCT1-high parent-gene phosphosite enrichment model embedded in recurrent kidney remodeling.